Sowmya Nadipineni

SOFTWARE DEVELOPER

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SUMMARY

- Software Developer with **4+ years** of experience building **backend, full-stack**, and **cloud-native solutions** across mobility, education, and enterprise domains.
- Proficient in **Java (Spring Boot, Hibernate, JPA)** and **Python (Flask, Django, FastAPI), delivering 20+ RESTful microservices** with high scalability and maintainability.
- Improved **API and database** performance by 55% through optimized queries, indexing, and caching strategies in **PostgreSQL** and **MySQL.**
- Designed and deployed 14+ responsive **UI components** with **React.js** and **TypeScript**, enhancing usability and accessibility for 25K+ users.
- Automated **CI/CD pipelines** using **GitHub** Actions, **Jenkins**, and **Azure DevOps**, reducing deployment effort by 6+ hours per release cycle.
- Skilled in cloud infrastructure (**AWS, GCP, Kubernetes, Docker, Terraform**) with proven success containerizing applications and scaling distributed systems.
- Experienced in **machine learning & AI (TensorFlow, Hugging Face, LangChain, AWS SageMaker)** for **NLP**, sentiment analysis, and computer vision applications.
- Integrated **Datadog** and **Grafana** for monitoring, enabling proactive alerting and reducing incident detection time by 35%.
- Strong foundation in OOP, algorithms, and distributed systems, with experience leading Agile/Scrum teams through successful feature delivery.

TECHNICAL SKILLS

Languages & Programming: Java (Spring Boot, Spring MVC, Hibernate, JPA), Python (Flask, Django, FastAPI), Golang, C, C++, JavaScript (ES6+), TypeScript, R, SQL
Frameworks & Libraries: React.js, Angular, Apache Spark, TensorFlow, Hugging Face Transformers, LangChain, Bootstrap, SASS, jQuery, OpenCV, Apache Mahout
Backend & APIs: RESTful APIs, Node.js, Express.js, oData Services, Swagger UI, Postman
Databases: PostgreSQL, MySQL, MongoDB, Redis, Cassandra, Elasticsearch, CosmosDB, NoSQL
Cloud, DevOps & CI/CD: AWS (EC2, S3, Lambda, SageMaker), GCP (Cloud Run, Kubernetes), Azure DevOps, Docker, Terraform, Jenkins, GitHub Actions, GitLab, Git
Monitoring & Observability: Datadog, Grafana, Logging Pipelines, Proactive Alerting
Enterprise & Integration: SAP, SAPUI5, ETL Pipelines, Distributed Systems
Machine Learning & AI: Decision Trees, CNNs, NLP Models, LLMs, Pinecone
Testing & QA: JUnit, Mockito, Unit Testing, TDD
Design & Collaboration: Figma, Responsive Web Design, Agile/Scrum, Jira
Core CS Concepts: Object-Oriented Programming (OOP), Algorithms, Data Structures

EDUCATION

University of New Haven — **West Haven, CT**
Master of Science in Computer Science — Jan 2022 – Dec 2023

Jawaharlal Nehru Technological University — **Andhra Pradesh, India**
Bachelor of Technology in Computer Science and Engineering — Jun 2017 – Apr 2021

PROFESSIONAL EXPERIENCE

Uber — **NY, USA**
Software Developer — **May 2024 – Current**

- Engineered a secure trip and payment management module using Java (Spring Boot + Hibernate) and Python (Flask), improving transaction throughput and reducing trip completion latency by 2.5 minutes.
- Built and deployed RESTful microservices for authentication, rider/driver profiles, and reporting workflows, scaling across 3 core mobility and delivery platforms.
- Designed 14+ responsive UI components with React.js and TypeScript, improving usability and navigation for 25,000+ monthly drivers and riders on the portal.
- Automated data ingestion workflows on AWS (S3, Lambda) with Docker and Azure DevOps CI/CD, reducing manual deployment effort by 6 hours per release cycle.
- Optimized PostgreSQL queries and indexing, cutting dashboard load times from 4.2s to under 1.8s, enhancing real-time tracking and reporting.
- Leveraged GitHub + GitLab workflows with branching strategies and peer reviews, reducing merge conflicts by 40% and accelerating production release readiness.
- Partnered with security teams to integrate JWT authentication and data encryption, ensuring compliance across all mobility and payments modules.
- Deployed FastAPI-based analytics services, integrated with Datadog and Grafana, enabling proactive monitoring and reducing

incident detection time by 35%.
- Orchestrated Agile sprint planning with Jira, coordinating backend, frontend, and DevOps teams to deliver 5 major feature releases in 6 months.

University of New Haven – International Enrollment Operations **CT, USA**
Software Developer **Jan 2022 –Dec 2023**

- Designed and implemented a full-stack course registration scheduler using Python (Flask), React.js, and PostgreSQL, applying Test-Driven Development (TDD) to deliver a stable solution supporting 500+ students during registration cycles.
- Streamlined enrollment data processing by automating SQL-based workflows in Slate CRM, eliminating over 12 hours of manual updates and enabling staff to process 200+ student records per batch.
- Built a Pandas-driven ETL pipeline incorporated with Excel macros for data cleaning, transformation, and validation, reducing dataset preparation time from 3 hours to 15 minutes for reporting.
- Collaborated with academic and IT teams to integrate Git version control and Jenkins CI/CD pipelines, achieving deployment readiness within 30 minutes for new feature releases.
- Enhanced admission analytics by developing custom Python scripts for statistical modeling, producing actionable reports that guided enrollment strategies for 5 academic terms.

Adani **India**
Software Engineer **Jan 2021 – Dec 2021**

- **Developed** a CNN-based image classification model using **TensorFlow** and **Keras**, achieving 65% accuracy on handwritten digit recognition with custom data augmentation techniques.
- **Engineered** preprocessing pipelines in **Python**, integrating **OpenCV** for edge detection and **Scikit-learn** for dimensionality reduction, reducing model training time for 50,000+ images.
- **Implemented** hyperparameter tuning workflows and trained deep learning models using **AWS SageMaker**, accelerating experiment cycles across multiple configurations.
- Built **Java**-based **RESTful APIs** for integrating machine learning inference services, improving scalability and enabling multi-client access to deployed models.
- Developed backend modules in **Java (Spring Boot + Hibernate)** for handling user authentication and data persistence, ensuring seamless integration with existing enterprise systems.
- **Deployed** model evaluation frameworks using **Flask** and **FastAPI**, enabling rapid testing and validation of CNN architectures across structured test sets.
- **Utilized Git** for version control and collaborated through weekly code reviews, ensuring consistent model updates and reproducible experimentation.
- **Visualized** learning curves, confusion matrices, and feature maps using **Matplotlib** and **Seaborn**, translating model performance insights into actionable feedback.
- **Explored** alternative models including **decision trees** and ensemble methods for benchmark comparison, ensuring robustness of selected architecture.
- **Documented** research methodology and findings, delivering technical reports to senior data scientists and aiding in integration planning for enterprise AI initiatives.

PROJECTS

Library Management System

- Built a full-stack platform with Flask, SQLite, and REST APIs for book lending/search, optimizing queries and UI to cut lookup time from 3.5s to 1.2s and handle 2,000+ transactions with 50+ concurrent users.

Noughts & Crosses AI Engine

- Developed an AI-powered game engine with Minimax and Alpha-Beta pruning, integrated into a Flask web interface supporting three difficulty levels and <50ms move generation for 200+ beta users.

Sentiment Analysis Optimizer

- Designed a sentiment analysis pipeline with scikit-learn, spaCy, and Transformers, deployed as a Flask REST API processing 5K+ live reviews weekly with <120ms inference speed.

Robotic automation in H&M company using data mining

- Applied ML models (Random Forest, Decision Trees, MLP) on 100K+ sustainability records to forecast transparency metrics (85% accuracy) and automated store workflows, saving 1,200 staff hours and $1.2M annually.

Personalized POI Recommendation (Java)

- Implemented a Spark-based recommendation system with collaborative filtering on 1M+ POI records, improving accuracy to 90% and cutting batch processing time by 3 hours for real-time personalization to 10K+ users.